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Addressing Data Imbalance in Medical Imaging: A Simulation Study Using Ensemble Transfer Learning for Skin Cancer Detection

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Abstract

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Abstract:

Data imbalance is a critical challenge in classification tasks, especially in the medical field, where accurate disease identification is essential. This study investigates the impact of imbalance in medical image datasets and proposes the use of transfer learning models combined with resampling and data augmentation to improve skin tumor detection. Images from ISIC (International Skin Imaging Collaboration) 2016, 2017, and 2018 were used, and simulations were performed varying the proportion of the minority class between 10% and 50%, considering data augmentation and resampling, we used a threshold-based ensemble of 13 convolutional neural network models. We evaluated the performance of combinations with standardized metrics, placing emphasis on the  $F_2$ -score. The results show that individual models have difficulty identifying malignant tumors when the imbalance is higher (10% malignant and 90% benign samples). However, the use of resampling and data augmentation and ensemble improves the outcome of these models to predict the malignant class. We conclude that the combination of these techniques is effective in dealing with imbalance, providing improvements in predictive performance.

| Simulation   | Benign (%) | Malignant (%) |
|--------------|------------|---------------|
| Simulation 1 | 90         | 10            |
| Simulation 2 | 80         | 20            |
| Simulation 3 | 70         | 30            |
| Simulation 4 | 60         | 40            |
| Simulation 5 | 50         | 50            |

Simulations used - percentage of malignant and benign images

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